

Project Design Phase Solution Architecture

Date	28 June 2026
Team ID	23A51A4229
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

The Metro Ticket Booking System is developed on the ServiceNow platform using a layered architecture that separates the user interface, business processing, and data management components. Passengers can book metro tickets through a Service Catalog form by providing travel details such as the departure station, destination station, travel date, and number of passengers. The system processes the booking request by fetching fare details from the metro fare database and computing the total ticket amount automatically using Catalog Client Scripts, GlideAjax, and Script Includes.

- **Presentation Layer:** The Service Catalog offers a simple self-service portal where users can enter their travel information, including source station, destination station, journey date, and passenger count to reserve metro tickets.
- **Application Logic Layer:** Catalog Client Scripts, GlideAjax, and Script Includes validate user inputs, obtain fare details from the database, and calculate the ticket price automatically based on the selected route and number of passengers.
- **Data and Service Layer:** ServiceNow database tables maintain information related to metro stations, fare structures, and ticket bookings. Additional services such as notifications, QR code generation, and dashboards support ticket delivery, user communication, and reporting with analytical insights.

Solution Architecture Diagram:

